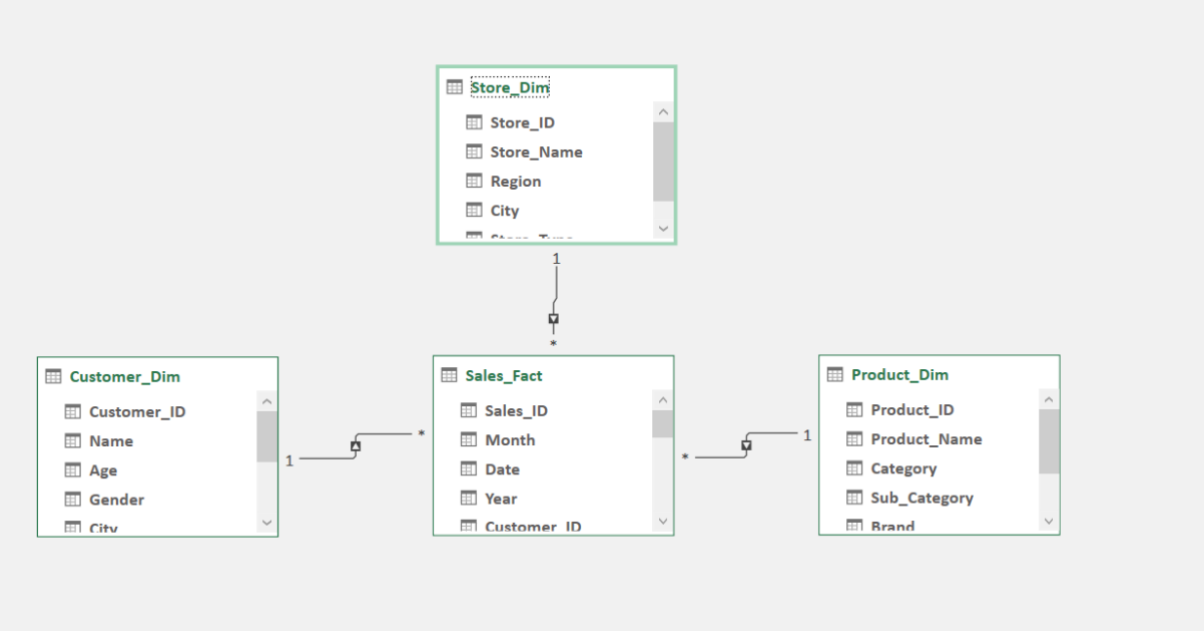
E-commerce Dataset Analysis

Project Objective:

To develop Business Intelligent dashboard providing actionable insights to sales performance. The project aims to establish relationships between sales facts (fact table) and dimensional data (Customer, Product, and Store) to answer critical business questions.

Dataset Description:

This is an E-commerce transaction dataset collected over 2023-2024. It comprises of Sales Facts table, followed by dimensional tables providing insights to customer demographics, product inventory, and store locations. Following explains how connection is made: Sales_Fact is the core datasheet. It is connected to the Customer_Dim with Customer_ID; Product_Dim with Product_ID; and Store_Dim with Store_ID. The dimensional sheets have necessary details required to answer business queries, such as membership, categories, costs, store types, etc. Processed dataset using Excel via Power Query for data cleaning and modelling. The dataset was trimmed and cleaned, null values replaced, duplicates removed, and necessary columns were engineered to facilitate smooth analysis.



Column Description:

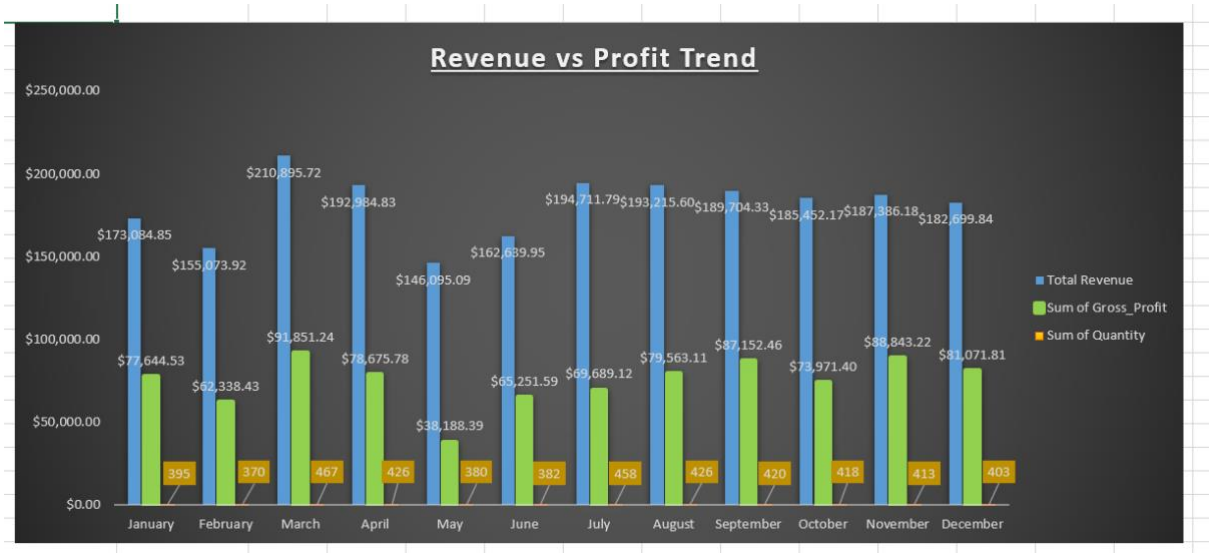
Column Name	Description
Sales_ID	Unique identifier for each transaction.
Total_Amount	Final revenue generated after discounts are applied.
Quantity	Number of units sold in a single transaction.
Gross_Profit	Monetary gain or loss from a sale (Revenue - Cost).
Order_Date (or Month/Year)	Used for time-series analysis.
Category/Sub-Category	Detailed product groupings.
Region	Geographical location.
Store_Type	Channel used for the sale (Online, Flagship, Outlet).
Loyalty_Level	Customer segments (Platinum, Gold, Silver, Bronze) based on value.
Profit_Status	(Profit/Loss) used for diagnostic analysis.
Profit_Margin:	Percentage of revenue that is profit (Gross Profit/Total Amount).
Discount_Value	Amount saved by the customer.

Insights Arrived:

Revenue & Volume: Total Revenue: $2,173,944.27

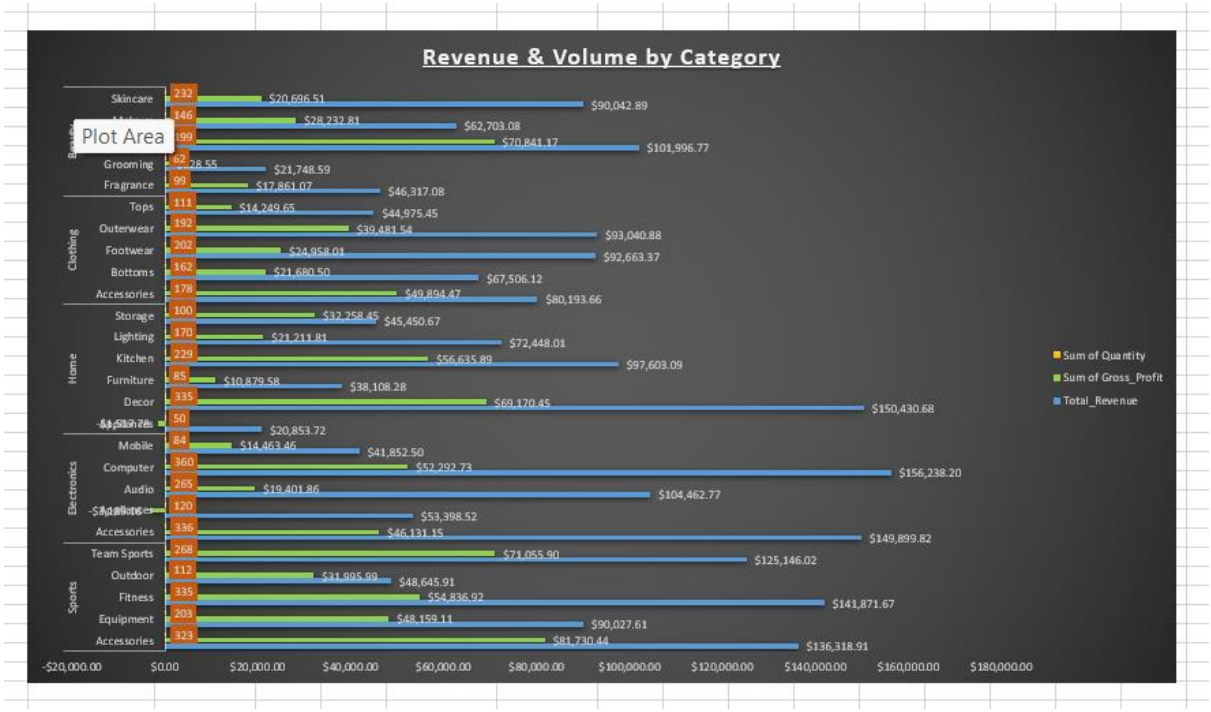
Total Quantity Sold: 4,958

Total Profit: $894,241.08.



Product Performance: Top 5 Sellers by Brand: Cummings and Sons; French Ltd; Hogan-Lee; Lucero LLC; Smith-Charles

Top 5 Sellers by Sub-Categories: Accessories; Computer; Décor; Fitness; Team Sports

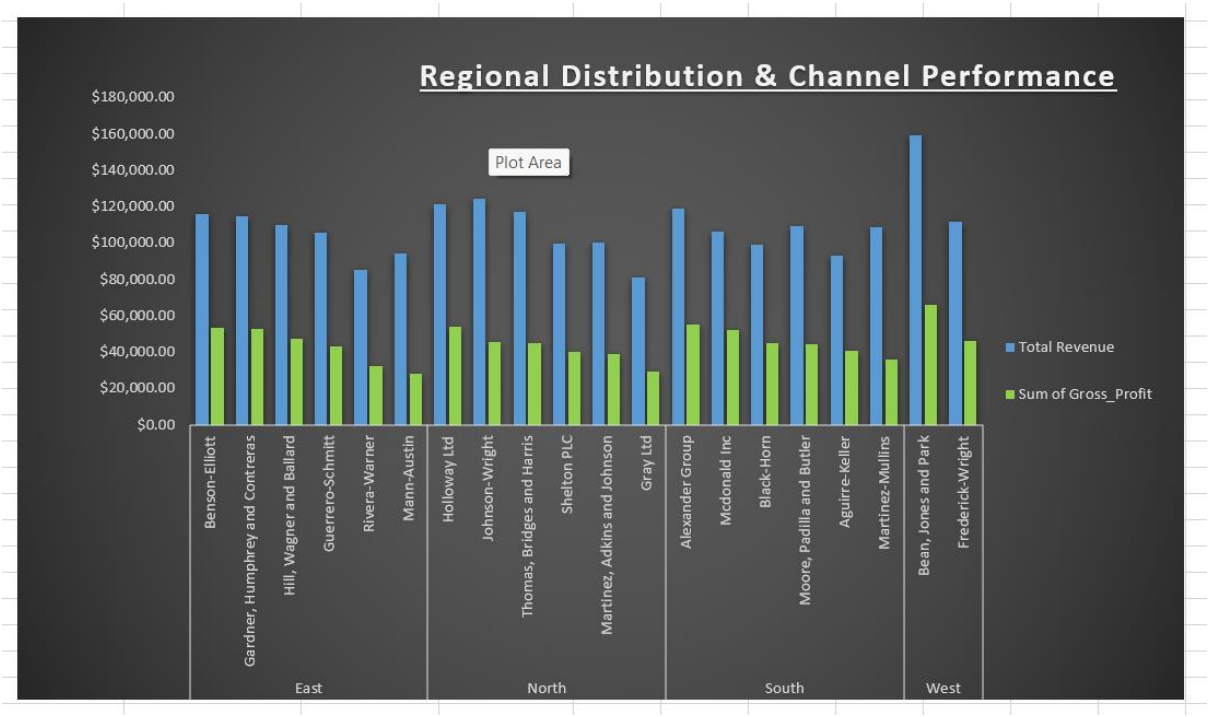


Geographic Trends: Region Generating Highest Sales Volume: North: Total Revenue $643,578.64.

Top 5 States Generating Highest Sales Volume [Region-wise]:

Region	State	Total Revenue
East	Minnesota	$20,295.34

	Montana	$22,323.66
	New York	$21,372.49
	Oklahoma	$18,703.96
	Vermont	$18,290.83
North	Florida	$23,291.27
	New Hampshire	$23,725.48
	New Jersey	$20,219.37
	Vermont	$21,988.75
	Wisconsin	$21,072.57
South	Connecticut	$25,954.50
	Iowa	$21,537.33
	New Hampshire	$22,331.31
	New Jersey	$26,820.49
	Oklahoma	$21,552.28
West	Alaska	$11,912.09
	Indiana	$9,296.45
	Kansas	$16,775.43
	New York	$11,418.05
	Tennessee	$9,265.46



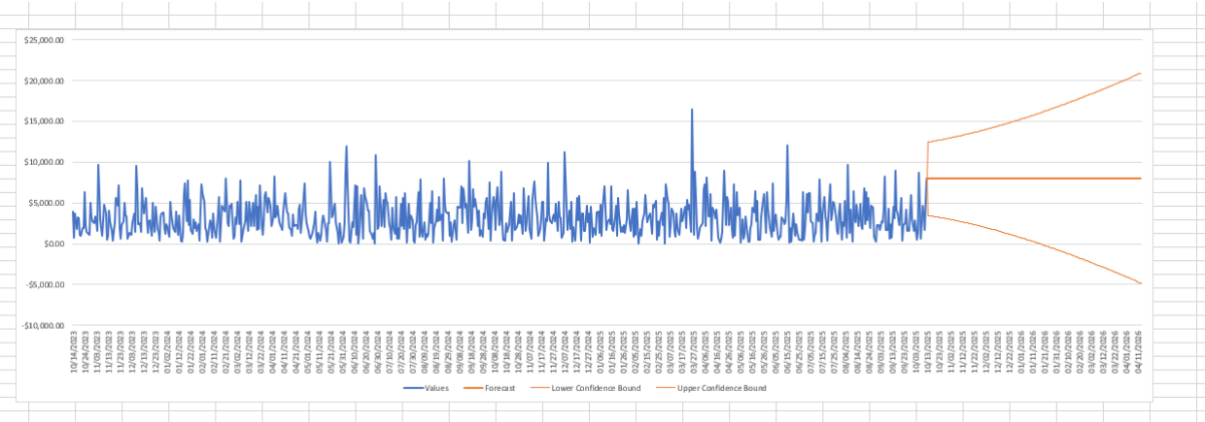
<u>Store Comparison</u>: Revenue generated from online stores is higher than compared to Flagship. Flagship generate higher revenue than outlets.

Customer Demographics: Age Distribution: Adult 199; Young Adult 129; Elderly 172.

Gender Distribution: Male 283; Female 262.

Profit Leakage: Some transactions have a negative profit margin primarily driven by High Discounts exceeding the 20% threshold.

Sales Forecasting: Based on the Order_Date trends, projected sales consecutive months.

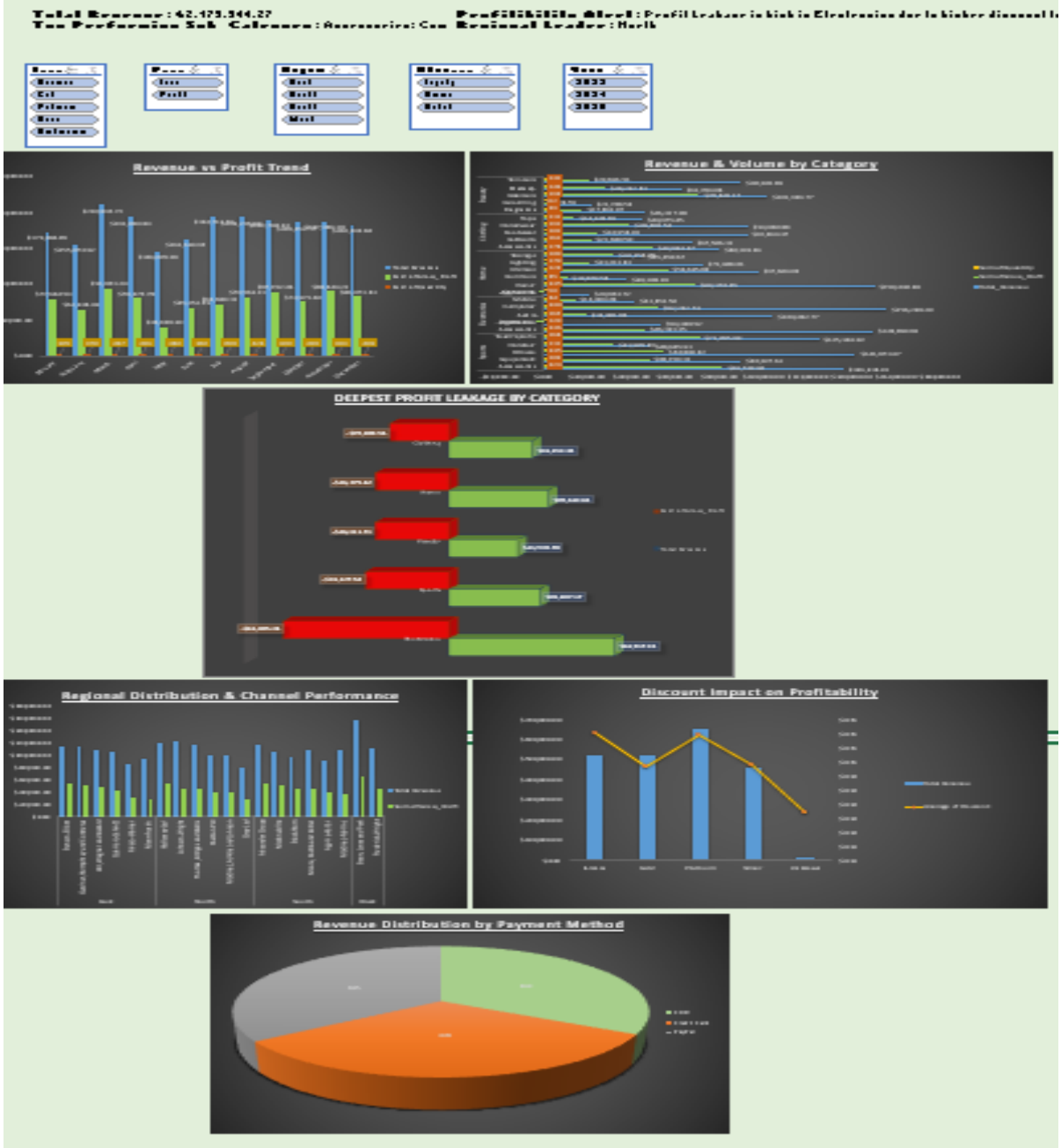


Pricing Strategy: Data shows that any discount exceeding 12% results in a negative net profit. Quantity sold for these items does not increase significantly with deeper discounts. Sticking with less than 10% discount would work without a significant drop in total sales volume.

Operational Improvement: Mode of payment are mostly online and COD. Could give offers on credit cards.

Conclusion:

The business generated a total revenue of $2,173,944.27 from 4,958 units sold, resulting in a total profit of $894,241.08. The North region led in sales volume with a total revenue of $643,578.64. Online stores outperformed flagship stores and outlet stores. Analysis reveals that any discount exceeding 12% typically results in a negative net profit. Maintaining discounts below 10% is recommended. Customer demographics is relatively balanced by gender. The largest sales segment is Adults, followed by elderly and young adults. Most of the transactions are completed through online and COD method. Recommended credit card offers.


Revenue vs Profit Trend
Revenue & Volume by Category
DEEPEST PROFIT LEAKAGE BY CATEGORY
Regional Distribution & Channel Performance
Discount Impact on Profitability
Revenue Distribution by Payment Method